

```
In [44]: import pandas as pd
```

```
In [85]: emp = pd.read_excel(r"C:\Users\ANIL\Downloads\Rawdata.xlsx")
```

```
In [86]: emp
```

```
Out[86]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

Data Cleaning

```
In [126... emp['Name']
```

```
Out[126... 0    Mike
1    Teddy
2    Umar
3    Jane
4    Uttam
5    Kim
Name: Name, dtype: object
```

```
In [127... emp['Name'] = emp['Name'].str.replace(r'\W', '', regex=True)
```

```
In [128... emp['Name']
```

```
Out[128... 0    Mike
1    Teddy
2    Umar
3    Jane
4    Uttam
5    Kim
Name: Name, dtype: object
```

```
In [50]: emp.columns
```

```
Out[50]: Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
In [51]: emp.head(1)
```

Out[51]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+

In [52]: emp['Domain']

Out[52]:

0	Datascience#\$
1	Testing
2	Dataanalyst^^#
3	Ana^alytics
4	Statistics
5	NLP

Name: Domain, dtype: object

In [108... emp['Domain'] = emp['Domain'].str.replace(r'\W', '', regex=True)

In [112... emp

Out[112...:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34 years	Mumbai	5000	2
1	Teddy	Testing	45' yr	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67-yr	NaN	30000	5
5	Kim	NLP	55yr	Delhi	60000	10

In [55]: emp['Location'] = emp['Location'].str.replace(r'\W', '', regex=True)

In [72]: emp

Out[72]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	^#	2+
1	Teddy	Testing	45	Bangalore	%%	<3
2	Umar	Dataanalyst	NaN	NaN	\$%	4> yrs
3	Jane	Analytics	NaN	Hyderbad	^	NaN
4	Uttam	Statistics	67	NaN	-	5+ year
5	Kim	NLP	55	Delhi	^\$	10+

In [58]: emp['Location']

```
Out[58]: 0      Mumbai
         1      Bangalore
         2         NaN
         3      Hyderbad
         4         NaN
         5      Delhi
         Name: Location, dtype: object
```

```
In [129... emp['Age'] = emp['Age'].str.replace(r'\W','',regex=True)
```

```
In [130... emp
```

```
Out[130... 
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34years	Mumbai	5000	2
1	Teddy	Testing	45yr	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67yr	NaN	30000	5
5	Kim	NLP	55yr	Delhi	60000	10

```
In [60]: emp['Age']
```

```
Out[60]: 0      34years
         1      45yr
         2         NaN
         3         NaN
         4      67yr
         5      55yr
         Name: Age, dtype: object
```

```
In [132... emp['Age'] = emp['Age'].str.extract('(\d+)')
```

```
In [62]: emp['Age']
```

```
Out[62]: 0      34
         1      45
         2      NaN
         3      NaN
         4      67
         5      55
         Name: Age, dtype: object
```

```
In [133... emp
```

Out[133...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [88]: emp['Salary'] = emp['Salary'].str.replace(r'\W', '', regex=True)
```

```
In [89]: emp['Salary']
```

```
Out[89]: 0    5000
1    10000
2    15000
3    20000
4    30000
5    60000
Name: Salary, dtype: object
```

In [134...

emp

Out[134...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [91]: emp['Exp'] = emp['Exp'].str.extract('(\d+)')
```

In [135...

emp

Out[135...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [136...

```
clean_data = emp.copy()
```

In [137...

```
clean_data
```

Out[137...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In []:

In [150...

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [151...

```
clean_data = emp.copy()
```

In [152...

```
clean_data
```

Out[152...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [142... clean_data.isnull().sum()
```

```
Out[142... Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64
```

```
In [153... clean_data['Age'] = clean_data['Age'].fillna(np.mean(pd.to_numeric(clean_data['Age']
```

```
In [55]: clean_data['Age']
```

```
Out[55]: 0      34
1      45
2     50.25
3     50.25
4      67
5      55
Name: Age, dtype: object
```

```
In [154... clean_data
```

```
Out[154...   Name  Domain  Age  Location  Salary  Exp
0  Mike  Datascience  34  Mumbai   5000    2
1  Teddy   Testing   45  Bangalore  10000    3
2  Umar  Dataanalyst  50.25    NaN  15000    4
3  Jane   Analytics  50.25  Hyderbad  20000  NaN
4  Uttam  Statistics   67    NaN  30000    5
5  Kim     NLP       55    Delhi  60000   10
```

```
In [ ]:
```

```
In [158... clean_data['Exp'] = clean_data['Exp'].fillna(np.mean(pd.to_numeric(clean_data['Exp']
```

```
In [159... clean_data['Exp']
```

```
Out[159... 0      2
1      3
2      4
3     4.8
4      5
5     10
Name: Exp, dtype: object
```

```
In [160... clean_data
```

Out[160...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	NaN	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [161...

```
clean_data['Location'].isnull().sum()
```

Out[161...

2

In [162...

```
clean_data['Location']
```

Out[162...

```
0      Mumbai
1    Bangalore
2         NaN
3    Hyderbad
4         NaN
5       Delhi
Name: Location, dtype: object
```

In [163...

```
clean_data['Location'] = clean_data['Location'].fillna(clean_data['Location'].mode(
```

In [164...

```
clean_data['Location']
```

Out[164...

```
0      Mumbai
1    Bangalore
2    Bangalore
3    Hyderbad
4    Bangalore
5       Delhi
Name: Location, dtype: object
```

In [165...

```
clean_data
```

Out[165...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [166... `clean_data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     object
3   Location    6 non-null     object
4   Salary      6 non-null     object
5   Exp         6 non-null     object
dtypes: object(6)
memory usage: 420.0+ bytes
```

In [167... `clean_data['Age'] = clean_data['Age'].astype(int)`

In [168... `clean_data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int32
3   Location    6 non-null     object
4   Salary      6 non-null     object
5   Exp         6 non-null     object
dtypes: int32(1), object(5)
memory usage: 396.0+ bytes
```

In [169... `clean_data['Salary'] = clean_data['Salary'].astype(int)`
`clean_data['Exp'] = clean_data['Exp'].astype(int)`

In [170... `clean_data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int32
3   Location    6 non-null     object
4   Salary      6 non-null     int32
5   Exp         6 non-null     int32
dtypes: int32(3), object(3)
memory usage: 348.0+ bytes
```



```
In [171... clean_data['Name'] = clean_data['Name'].astype('category')
clean_data['Domain'] = clean_data['Domain'].astype('category')
clean_data['Location'] = clean_data['Location'].astype('category')
```

```
In [172... clean_data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   Name        6 non-null      category
 1   Domain       6 non-null      category
 2   Age         6 non-null      int32
 3   Location    6 non-null      category
 4   Salary      6 non-null      int32
 5   Exp         6 non-null      int32
dtypes: category(3), int32(3)
memory usage: 866.0 bytes
```

```
In [173... clean_data
```

```
Out[173...
   Name   Domain   Age  Location  Salary  Exp
0  Mike  Datascience  34   Mumbai   5000   2
1  Teddy   Testing   45  Bangalore  10000  3
2  Umar  Dataanalyst   50  Bangalore  15000  4
3  Jane   Analytics   50  Hyderabad  20000  4
4  Uttam  Statistics   67  Bangalore  30000  5
5  Kim     NLP        55    Delhi   60000  10
```

```
In [174... clean_data.to_csv('clean_data.csv')
```

```
In [175... import os
os.getcwd()
```

```
Out[175... 'C:\\Users\\ANIL'
```

EDA TECHIQE APPLY

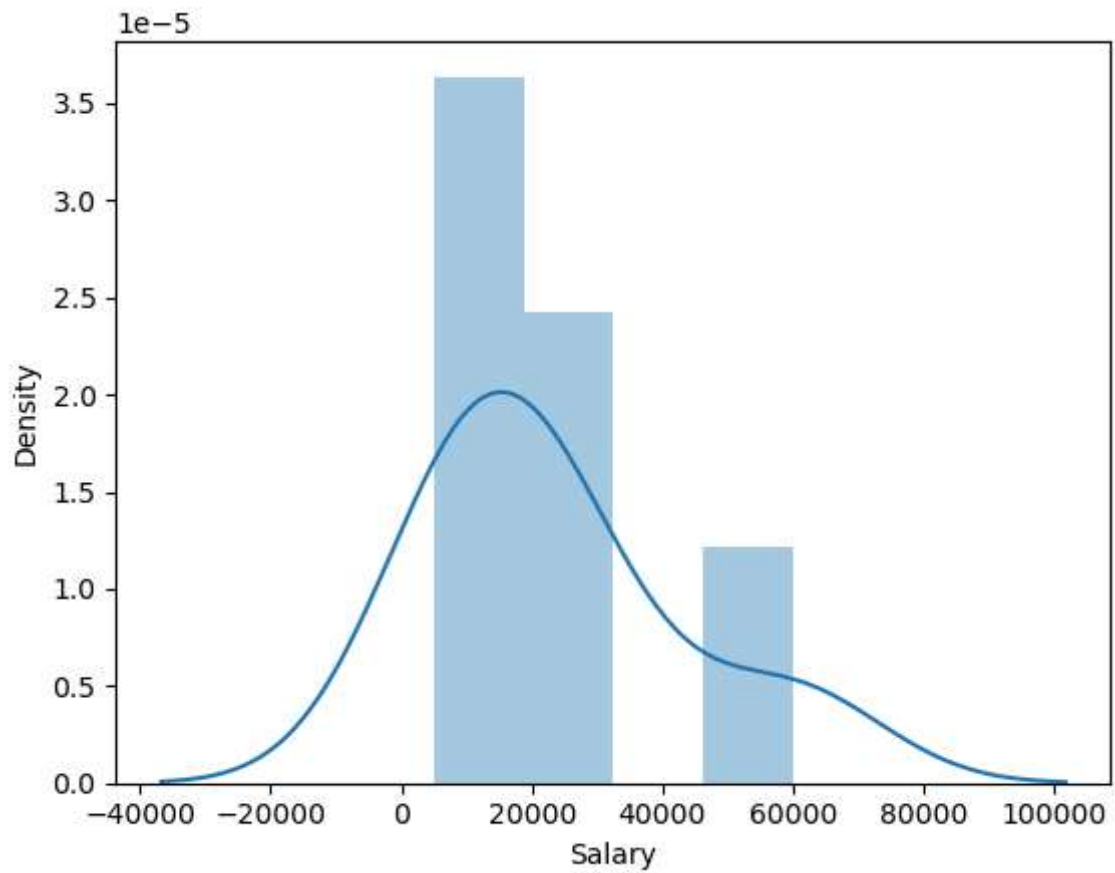
```
In [176... import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [177... import warnings
warnings.filterwarnings('ignore')
```

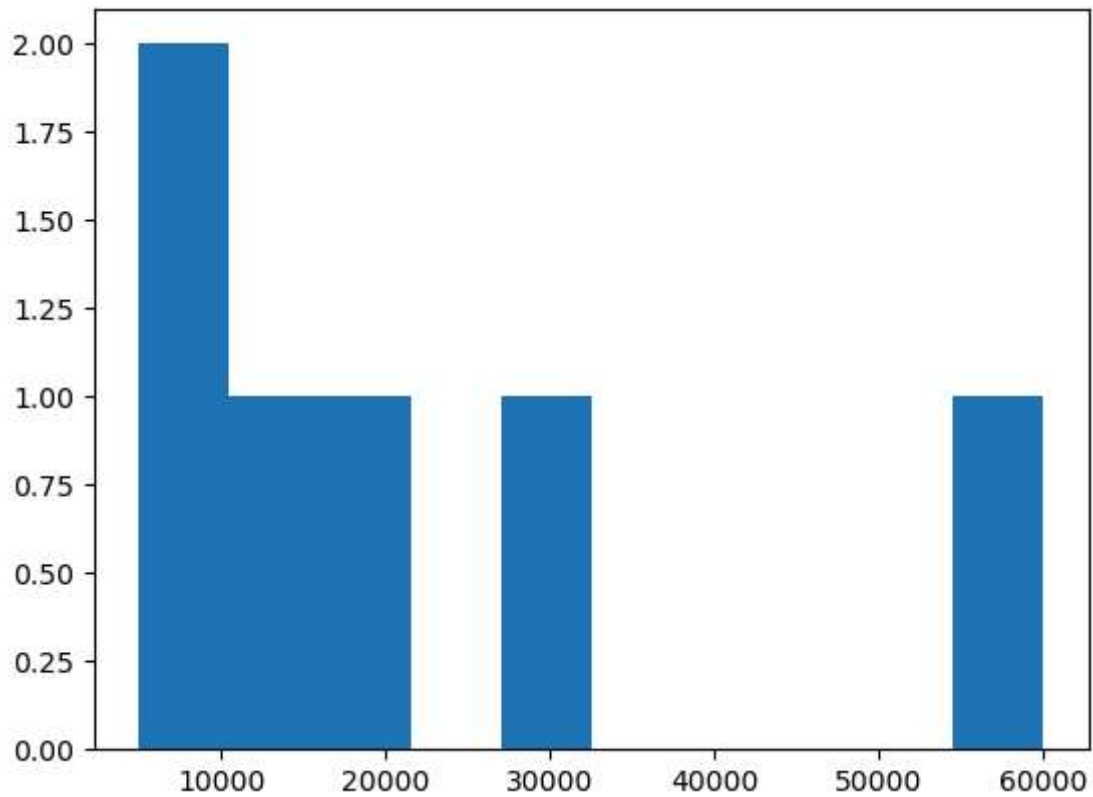
```
In [178... clean_data['Salary']
```

```
Out[178... 0    5000  
1   10000  
2   15000  
3   20000  
4   30000  
5   60000  
Name: Salary, dtype: int32
```

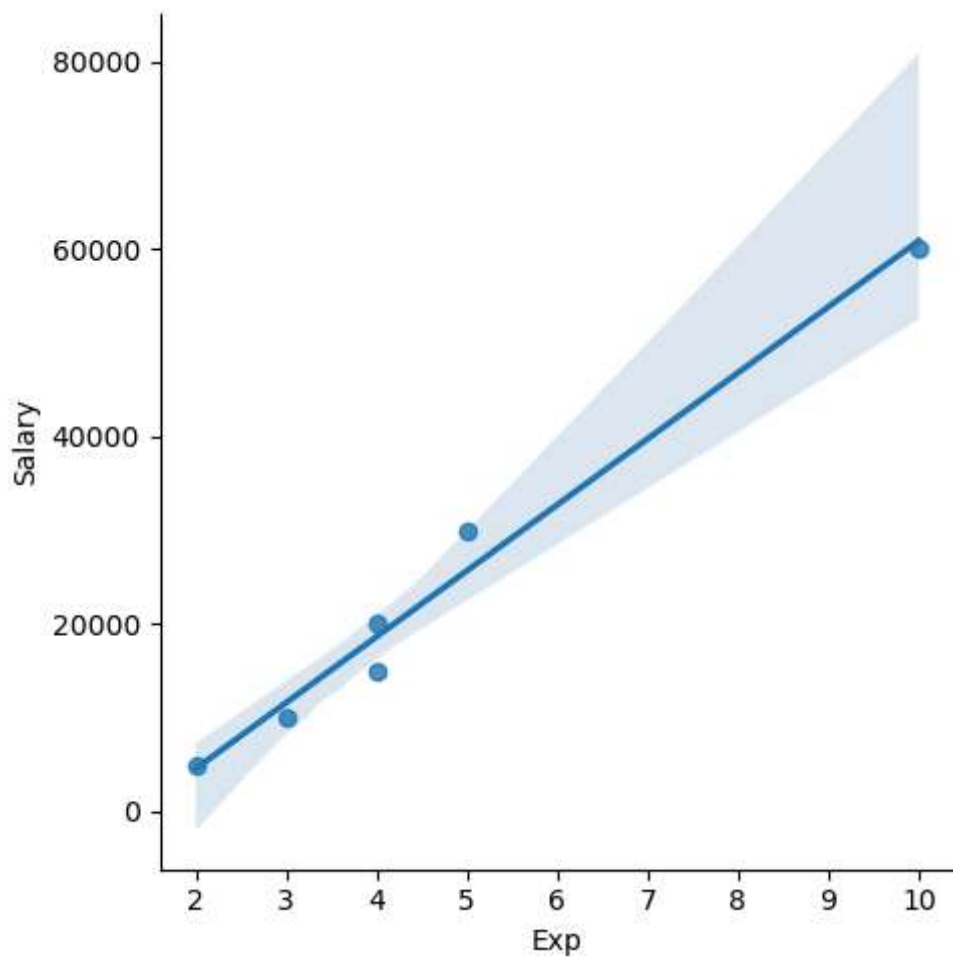
```
In [179... vis1 = sns.distplot(clean_data['Salary'])
```



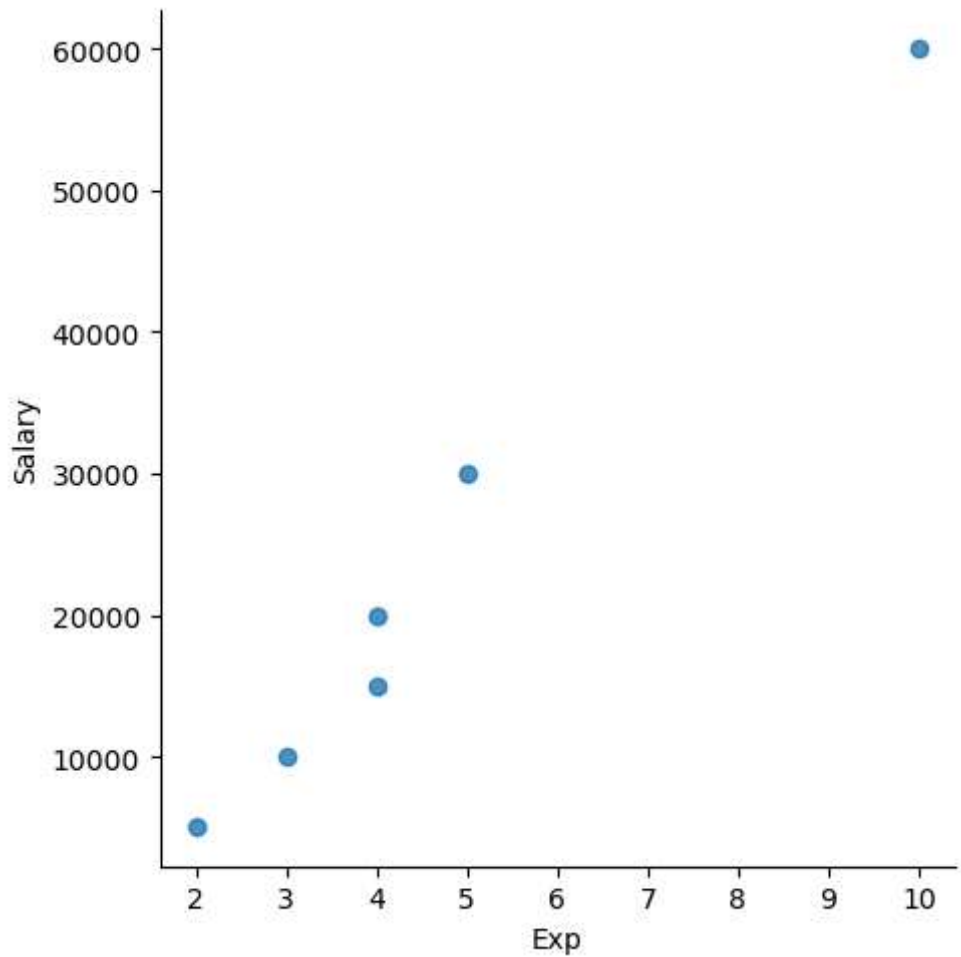
```
In [180... vis2 = plt.hist(clean_data['Salary'])
```



```
In [182... vis4 = sns.lmplot(data=clean_data,x = 'Exp', y='Salary')
```



```
In [183... vis5 = sns.lmplot(data=clean_data,x = 'Exp', y = 'Salary', fit_reg=False)
```



```
In [184... clean_data[:]
```

```
Out[184...
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [185... clean_data[0:6:2]
```

Out[185...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
2	Umar	Dataanalyst	50	Bangalore	15000	4
4	Uttam	Statistics	67	Bangalore	30000	5

In [186...

clean_data[::-1]

Out[186...

	Name	Domain	Age	Location	Salary	Exp
5	Kim	NLP	55	Delhi	60000	10
4	Uttam	Statistics	67	Bangalore	30000	5
3	Jane	Analytics	50	Hyderbad	20000	4
2	Umar	Dataanalyst	50	Bangalore	15000	4
1	Teddy	Testing	45	Bangalore	10000	3
0	Mike	Datascience	34	Mumbai	5000	2

In [187...

clean_data.columns

Out[187...

Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')

In [188...

x_iv = clean_data[['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp']]

In [189...

x_iv

Out[189...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [190...

y_dv = clean_data[['Salary']]

In [191...

y_dv

Out[191...

Salary

0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

In [192...

emp

Out[192...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [193...

clean_data

Out[193...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [194...

x_iv

Out[194...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Data science	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Data analyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderabad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [195...

y_dv

Out[195...

	Salary
0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

In [196...

clean_data

Out[196...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Data science	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Data analyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderabad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [197...

imputation = pd.get_dummies(clean_data, dtype=int)

In [199...

imputation

Out[199...

	Age	Salary	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar	Nan
0	34	5000	2	0	0	1	0	0	
1	45	10000	3	0	0	0	1	0	
2	50	15000	4	0	0	0	0	1	
3	50	20000	4	1	0	0	0	0	
4	67	30000	5	0	0	0	0	0	
5	55	60000	10	0	1	0	0	0	

In [200...

clean_data

Out[200...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderabad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [201...

imputation

Out[201...

	Age	Salary	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar	Nan
0	34	5000	2	0	0	1	0	0	
1	45	10000	3	0	0	0	1	0	
2	50	15000	4	0	0	0	0	1	
3	50	20000	4	1	0	0	0	0	
4	67	30000	5	0	0	0	0	0	
5	55	60000	10	0	1	0	0	0	

In [202...

```

from gtts import gTTS
from IPython.display import Audio

text_to_speech = gTTS('Hi All My Name is Ajay Adams Gangarapu i am working as Da

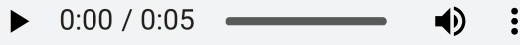
text_to_speech.save('text_to_speech_gtts.wav')
sound_file = 'text_to_speech_gtts.wav'

```



```
Audio(sound_file, autoplay=False)
```

Out[202...]



In []:

In []:

raw data with lot of regex, missing, unclean data

regex, clean

fill missing numerical & categorical

clean_dataset (data cleaning) 3 month - 5 month

outlier treatment, univariate, bivariate, correlation

split the data into x_iv & y_dv

impute categorical data to numerical

In []:

In []:

In []:

In []: